

K. Brenndörfer

Low-Noise-Amplifier optimised for minimum Noise Figure at 1.9 GHz using BFP420

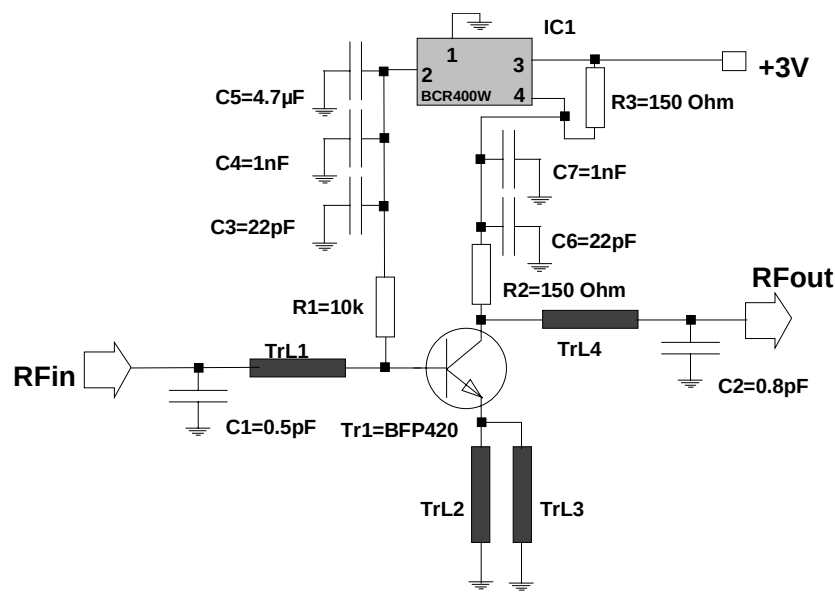
This application note provides general information, print layout and list of used components, circuit layout and measured data of a low noise amplifier at 1.9GHz using SIEMENS SIEGET[®] 25 BFP420. This circuit is optimised for low noise figure values.

Data at 1.9 GHz (3V and 5mA):

Gain=13 dB / IP3out=10dBm / NF=1.4dB / R_{Lin} >7dB / R_{Lout}>20 dB

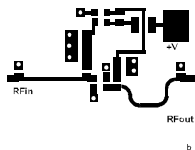
The amplifier application circuit has been optimised to achieve optimum noise figure and good stability at 1.9 GHz, combined with low operating current and voltage for use in handheld equipment.

The transistor is matched using microstrip lines at input and output in conjunction with capacitors to ground. The added emitter inductance reduces gain but it provides the best overall matching conditions for maximum gain and minimum noise figure. The active bias controller BCR400W is used to provide bias stability over the operating temperature range. For additional information on this part please refer to application note No.014. The collector resistor is used to increase stability of the RF-transistor and omits the need to use $\lambda/4$ transmission lines for biasing. In order to avoid AF oscillations of the BCR400W, the blocking capacitor at the base of the RF transistor should be at least ten times the value used at collector.

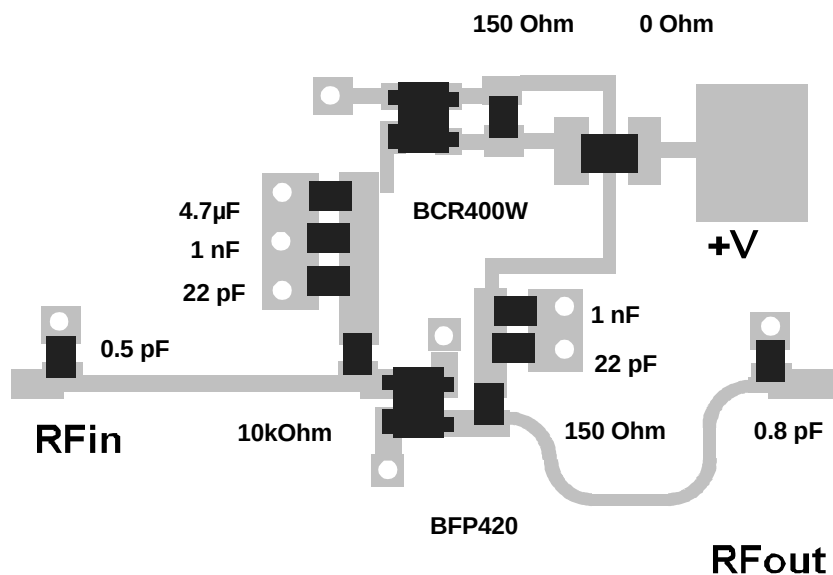


Component	Value	Unit	Comment
R1	10	k Ω	Bias 0603
R2	150	Ω	to improve stability and output return loss 0603
R3	150	Ω	bias resistor for bcr400w / collector current 0603
C1	0.5	pF	inputmatch 0603
C2	0.8	pF	outputmatch 0603
C3	22	pF	rf-short 0603
C4	1	nF	rf-short 0603
C5	4.7	μ F	
C6	10	nF	rf-short 0603
C7	22	pF	rf-short 0603
C8	1	nF	rf-short 0603
Tr1			SIEGET [®] BFP420 SOT343
TrL1			inputmatch
TrL2			emitter-microstrip-line
TrL3			emitter-microstrip-line
TrL4			outputmatch
IC1			BCR400W, active bias controller SOT343
Substrate	FR4		h=0.5mm, $\epsilon_r=4.5$

PCB Layout and Component Placement:

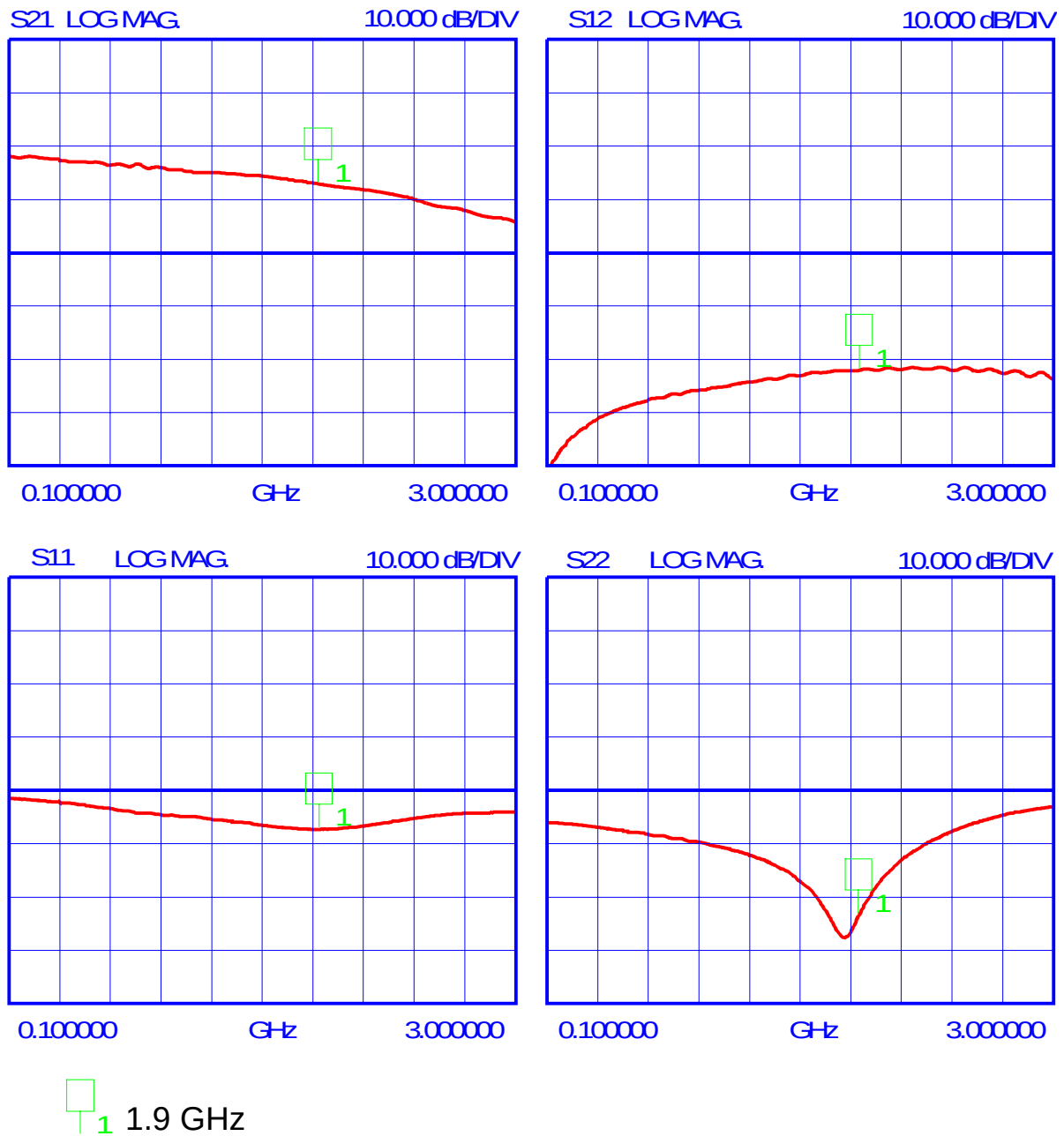


25mm x 20mm



b

Measurements:



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